14. IMPLANTATION AND AMNIOTIC CAVITY

DURATION : 14:52 STUDY SHEET

COURSE SUMMARY

This video explores the fascinating process of embryonic implantation and the formation of the primitive amniotic cavity. You will learn how the embryo, with its assimilative pole, penetrates the uterine lining thanks to specialised cells and aggressive syncytiotrophoblast activity, which generates micro-bleeding and prepares the ground for life. We will discuss cell differentiation between hypoblasts and epiblasts, which will respectively give rise to the digestive and neural systems. Ultimately, this session reveals the subtleties of the embryo's quest for embodiment, reinforcing an understanding of implantation challenges and their impact on successful pregnancy.

IMPLANTATION AND AMNIOTIC CAVITY

The **primitive embryo** has an **embryonic pole**, which is an assimilative pole. It always penetrates the **uterine lining** through this pole.

The uterine lining is shown here. The embryo displays **specific cells**, which capture information. This layer of cells forms what is called the **embryonic vesicle**.

The layer here is the **embryonic disc**. At the front, we have the cells of the **syncytiotrophoblast**. "Syncytio" indicates lipolytic activity, therefore an aggressive activity.

If the lining is not ready or if, and this is common, the embryo is not strong enough, **implantation** can fail. Often, during early miscarriages, the embryo lacks power, its metabolism is not strong enough. If the soul has not received this power, the embryo will not be able to enter the lining and will be naturally eliminated.

The syncytiotrophoblast releases an **acid** that attacks the lining. It's as if acid is being poured. The uterine system will then prepare to "welcome" the embryo into its "warm nest".

We will visualise this process through an animation and a more realistic film. This is a crucial moment, almost an incarnation, where the embryo agrees to "enter the flesh", into the maternal lining, into "the earth".

Very early on, there are **calcium-dependent cell junctions**. These adherent cells, called **cadherins** (calcium ions), provide cohesive strength.

When the embryo reaches the uterine lining, it has completed its journey. We observe a zone of concentrated cells, in front of which is a small layer of cells that change their name: the syncytiotrophoblast.

This aggressive lytic activity allows the cells to penetrate the lining by releasing an acid. This even causes **micro-bleeding** and a **micro-scar** in the uterus, at the implantation site. This phenomenon can sometimes lead to confusion regarding the dates of the last menstrual period.

The embryo always enters through the assimilative pole and in a perfectly straight manner. No deviation is tolerated, under penalty of implantation failure.

This penetration is made possible by a **powerful metabolic field**, an intense absorption that responds to a "will to live". It is a true quest for life, a "choice of the soul" to incarnate. This is why the embryo enters through the pole of the **micromeres**.

Let's now observe this image: we see the **blastocoel** and the cells of the embryonic disc. The cells of the syncytiotrophoblast, releasing a fluid, erode and penetrate the lining.

The uterine lining is rich, swollen, full of **blood vessels**, arteries, and small uterine glands, ready to welcome and nourish the **zygote**.

The embryo completely penetrates the lining. At the moment of its entry, a small cavity appears in the embryo: the **primitive amniotic cavity**.

Cells organise around this cavity. The cells facing the amniotic cavity are the **epiblastic cells**. The cells facing the blastocoel cavity are the future **hypoblastic cells**, of endodermal type.

In other words, some cells face the amniotic cavity and others face the blastocoel cavity. The latter will later participate in another formation, which we will explain.

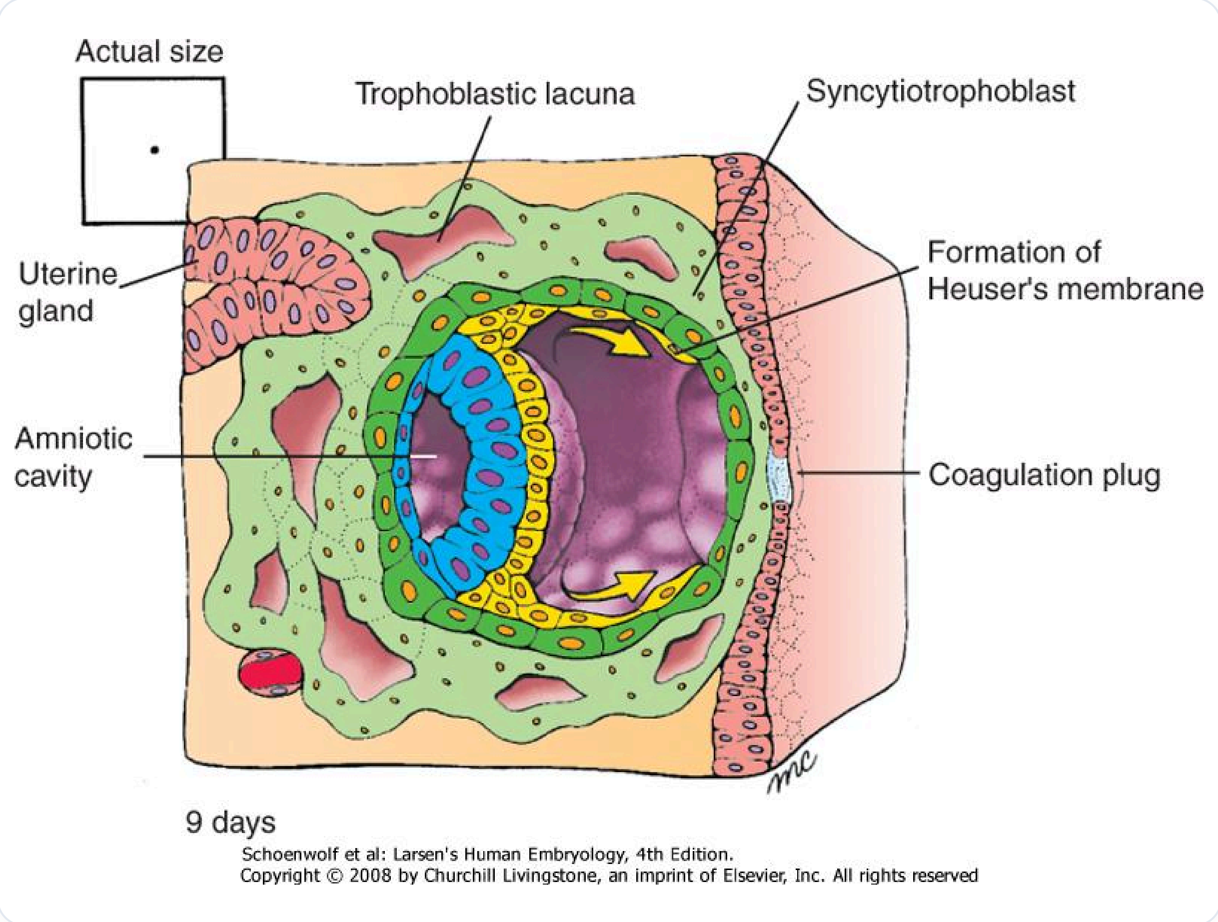


FIG. — Human embryo 9 days

It is important to imagine that at this stage, **cellular differentiation** is already appearing. Two cavities are forming:

- Everything that will be hypoblastic will give rise to the future **digestive system**.
- Everything that will be epiblastic will give rise to the future **neural system**.

Let's examine the mechanism more subtly. The embryo "is hungry", it needs nourishment. There is a continuous process of information exchange between the cells.

The cell penetrates the lining. It is constrained, it doesn't just slide. It uses acid to "push" and integrate itself.

The initial cells receive trophic information. The posterior cells are still nourished by the amniotic cavity of the blastocoel. But the central cells are compressed, lacking nourishment.

The membrane suddenly disappears, giving way to a **cellular exudate**. This exudate is the anlage of the amniotic cavity. The growth process is already underway.

What will this amniotic cavity become? It will become the "bag of waters", a primitive amniotic cavity that will completely envelop the embryo, surrounding it with a small liquid exudate.

In the end, the embryo swims and is bathed in this cavity. Cells, called **amnions**, will develop to nourish and grow this cavity.

The epiblastic cells will form the **nervous system** (the neural plate). Above this neural plate, we will later find the **primitive cerebrospinal fluid**, originating from the amniotic cavity.

This amniotic fluid, within the being, is so profound that we forget this liquid cavity within us. The **ventricular system** and this fluid at the centre of oneself are nothing other than amniotic fluid.

Outside the embryo, there is also amniotic fluid. The embryo at this stage drinks this fluid. It is completely informed by this pulsating, autocrine and paracrine fluid.

The information of **internal and external pressure** is crucial. The integration of the **coelom** (closure of the peritoneal cavity) and the closure of the **neural tube** will occur simultaneously. These are synchronicities, specific inductions.

Here, we still have a zygote. The **foetus** appears with the primitive axial process, i.e., **gastrulation**. It is then that there is a foetal form.

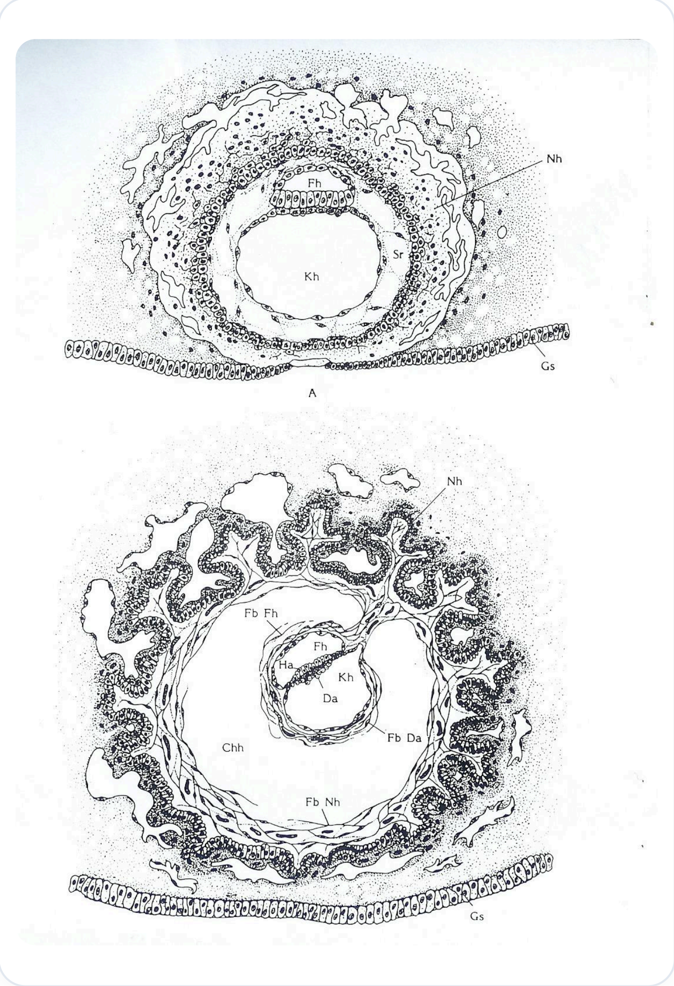


FIG. — Early embryonic development

In the meantime, the woman offers a "lunar landscape" rich in mucous membranes, glands, and nourishment. This is the moment of choice, where the embryo can penetrate the lining.

The embryo seeks to adhere, sending acid. There is very significant metabolic activity.

On the eighth day, the embryo penetrates the lining. Pressures to the right and left create the force necessary for the appearance of the amniotic cavity.

In the embryonic disc, at the moment this cavity appears, cells organise and orient themselves towards it. This is where the epiblast and hypoblast appear.

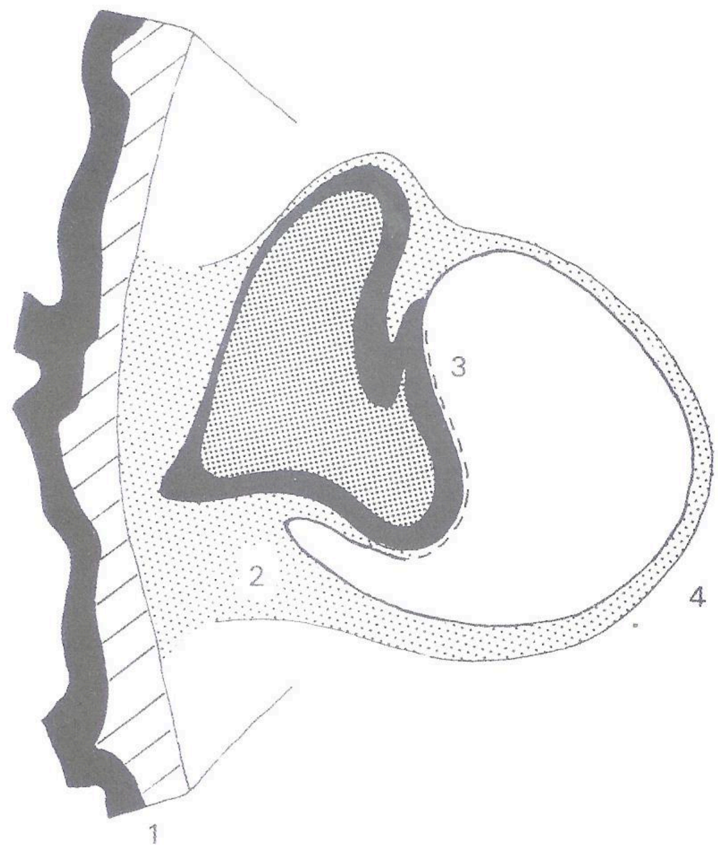


FIG. — Developing nephron

At this stage, we already have a preparation for digestive-type tissue and neurological-type tissue.

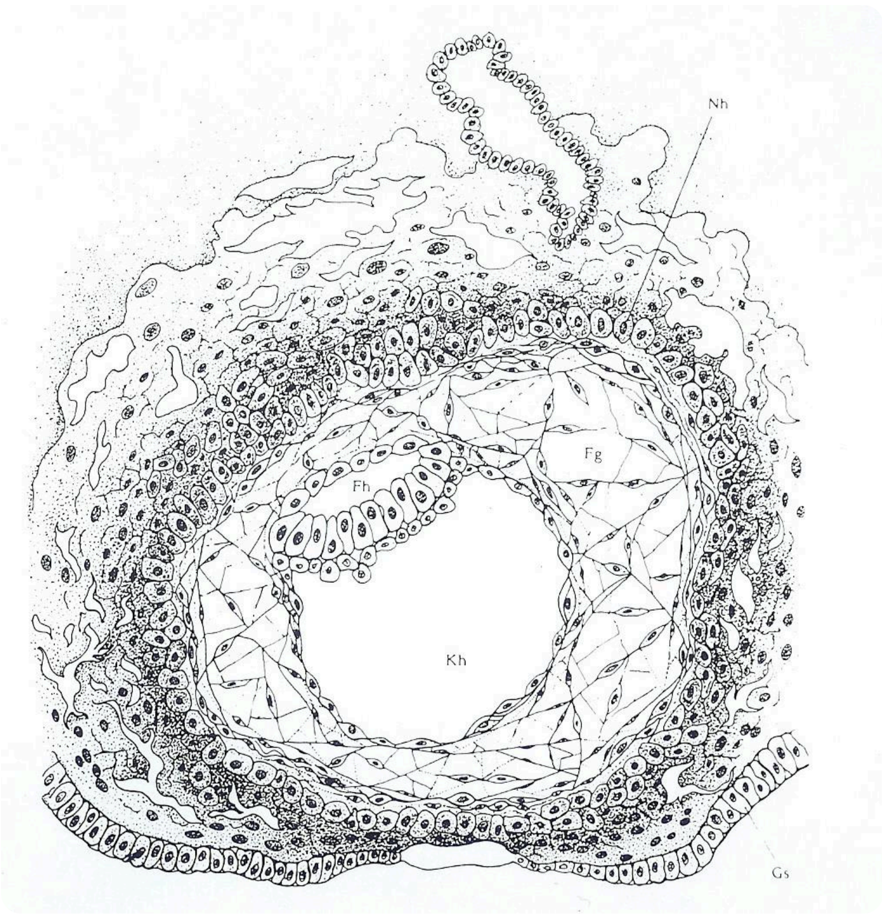


FIG. — Neurulation</td></tr></table>

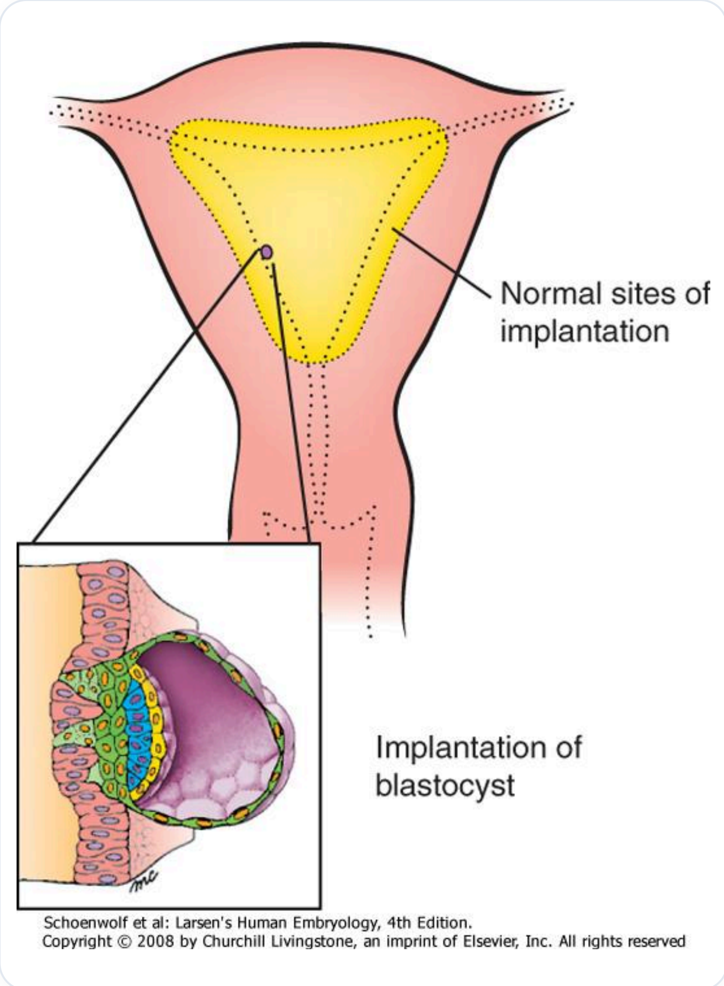
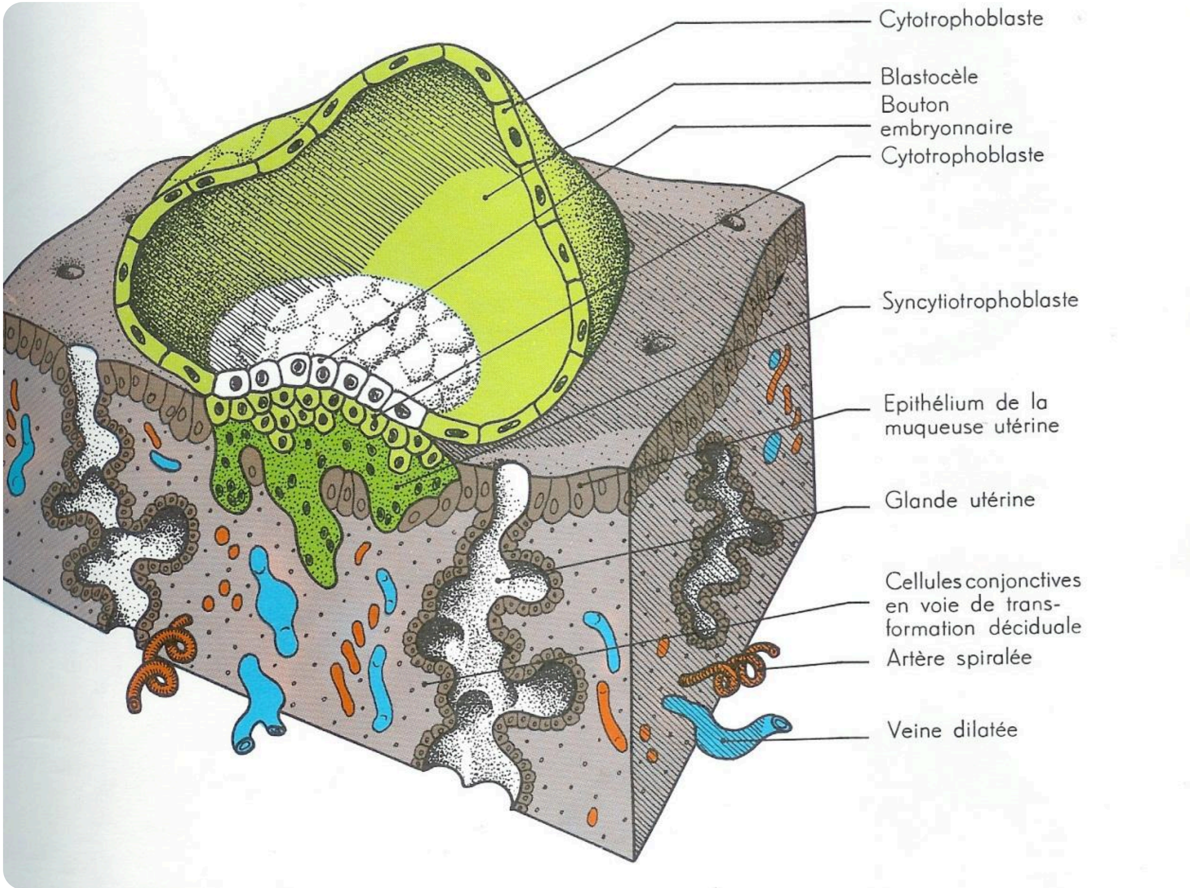


FIG. — Blastocyst implantation



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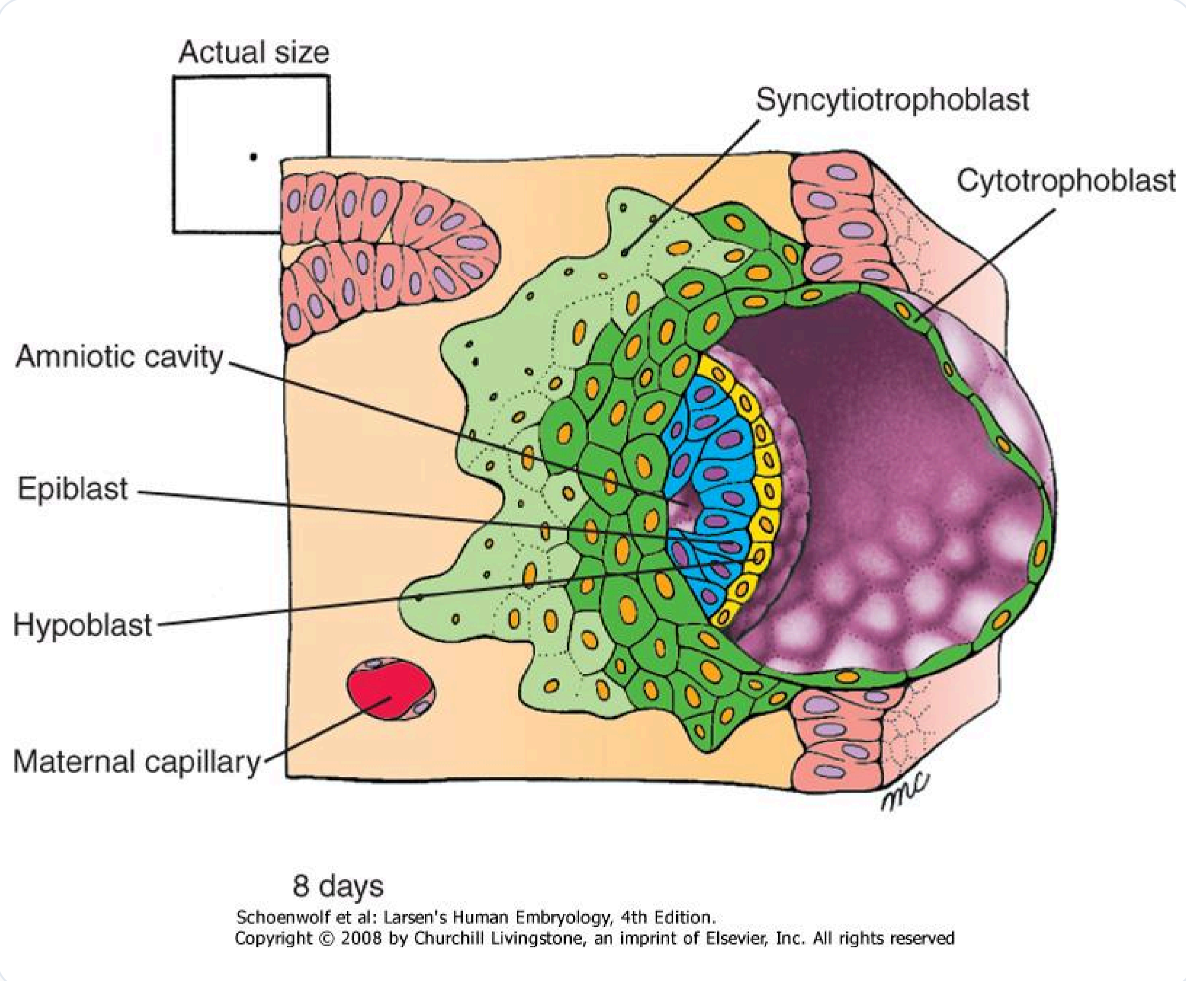

FIG. — 8th embryonic day

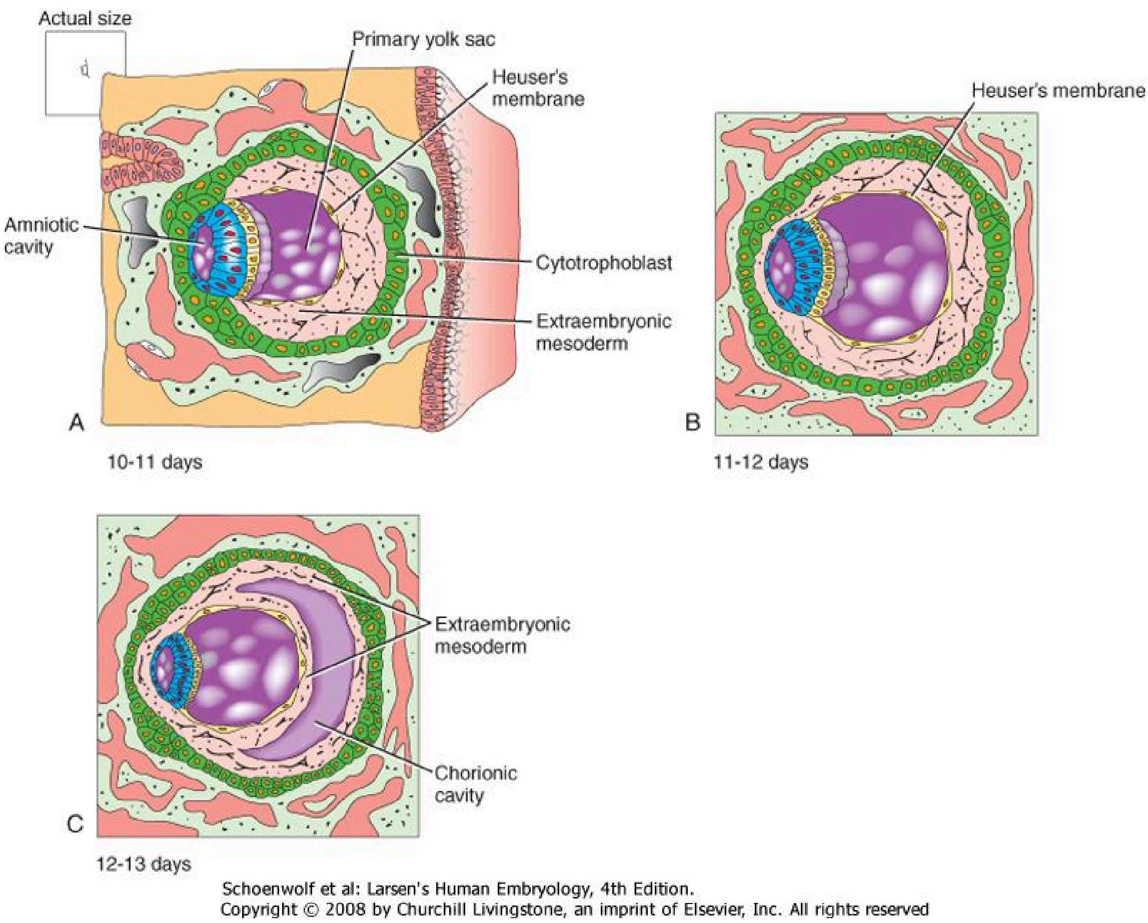

FIG. — Early embryonic development

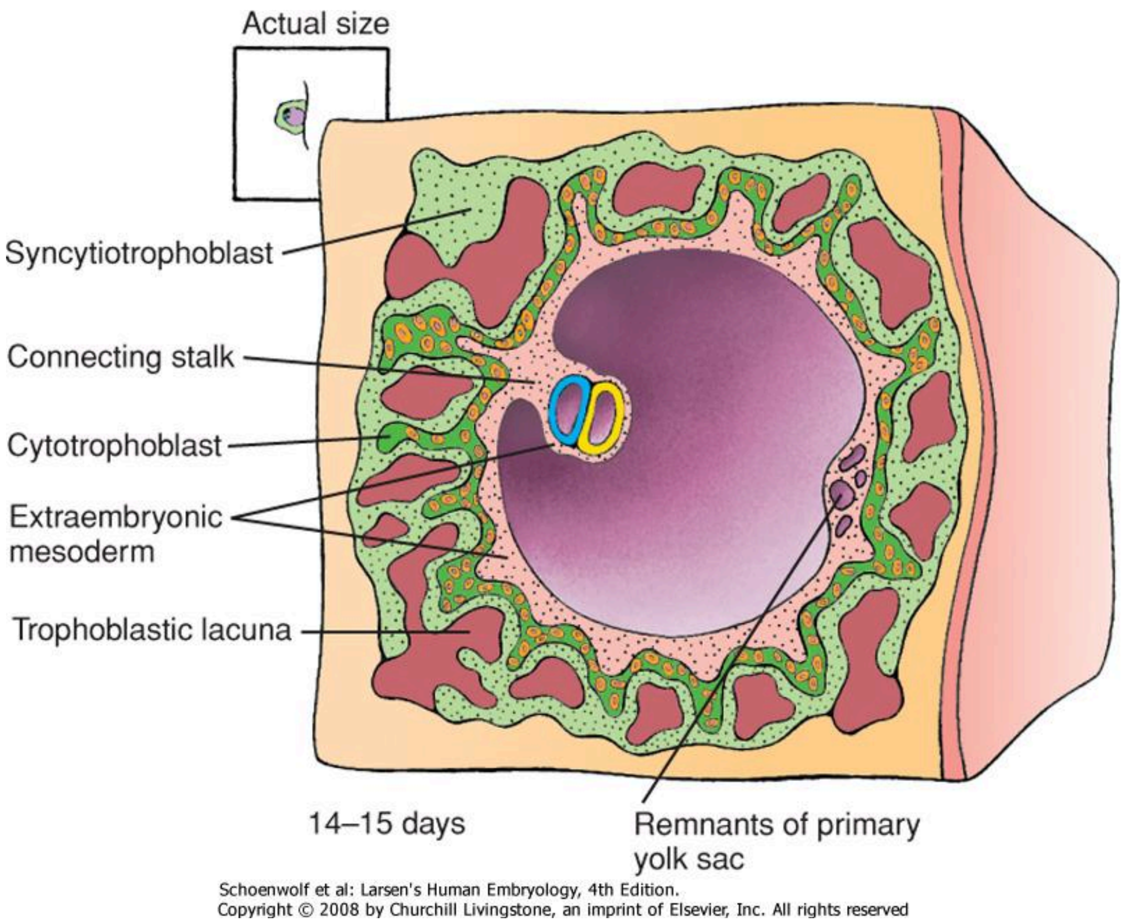

FIG. — Embryo 14-15 days